[uF]2Vec: Why Uninterpretable Features Lack CI-Projection

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ABSTRACT: The deletion or invisibility of uninterpretable features at the Conceptual-Intentional (CI) interface is usually treated as stipulative. This squib argues instead that these effects follow naturally from CI's vector-space architecture. Features are represented geometrically: interpretability corresponds to projection onto a CI-Potent Space, while uninterpretable features, transformed by non-linear feature checking, collapse into a Null Space and make no interpretive contribution. Form Copy, by contrast, preserves only linearly recoverable features and maps them onto the Potent Space. Modelling AGREE as an inter-dimensional mapping, the CI-Filter reduces to Null Space projection, unifying Deletion and Form Copy under a single architecture-driven principle.

Keywords: Interpretability, Feature, CI, Null Space, Modularity

1. Introduction

In the Minimalist Program, features are traditionally defined by three properties: interpretability (i/u), type (F), and value (Val). This tripartite architecture is active within the Narrow Syntax (NS) and reflected at the interfaces. Uninterpretable features ([uF]s) drive syntactic computation (Chomsky, 2007, 2008; Richards, 2007), pairing with their interpretable counterparts ([iF]s) for checking. Valuation applies post-checking, either within Narrow Syntax or externally (cf. Chomsky 2000, 2001; Kučerová 2018; Pesetsky & Torrego 2007). Interpretability further acts as a filter on Transfer. First, unchecked [uF]s cause a crash at the interfaces, as required by Full Interpretation (FI, cf. Chomsky, 1986, 1991). This is the FI-Filter. Second, only [iF]s are CI-visible; checked [uF]s are excluded. This CI-Filter has traditionally been implemented via Deletion (Chomsky, 1995), but this creates a *look-ahead* problem: Narrow Syntax cannot anticipate which features will induce crash at a later stage (Epstein et al., 1998; Epstein & Seely, 2002; Zeijlstra, 2014). Epstein, Kitahara and Seely (2010, 2012) address this by reconceiving Deletion as Invisibility: [uF]s simply do not register at CI. While this avoids the *look-ahead* problem, it preserves the stipulative assumption that checked [uF]s are semantically irrelevant. Yet the restriction itself plausibly reflects a core property of CI: only information recoverable under linearity is legible there. Existing accounts, however, merely redescribe this restriction instead of deriving it. The present Brief derives these filtering effects from the internal architecture of features, recast within a vectorial system, with the CI-Filter emerging as a consequence of Null Space projection.

2. From Featural to Vectorial Formalism

To quantify the CI-Filter, I recast the traditional feature system in vectorial terms.[1] This requires not merely representing features as vectors, but deriving central notions such as (un)intepretability and AGREE from their representations properties. On standard Minimalist assumptions, [uF]s must be checked against their [iF] counterpart (Chomsky, 2007, 2008; Richards, 2007). An alternative line of work treats [uF]s as unidentified features ([uX]s), whose identity is fixed only through interaction (Deal, 2015). Under Minimal Search (MS; cf. Ke 2019, 2022), AGREE proceeds as follows: a probing [uX] searches its domain, locates an [iF], establishes a dependency, and halts if fully satisfied; otherwise, probing continues until all relevant components are matched. The present proposal develops this intuition further in representational terms. [iF]s are directly legible at CI, while checked [~~u~~F]s are transformed by a non-linear, checking operation in such a way that they leave no CI-projection. The CI-Filter thus follows from the mechanics of the system itself, rather than from stipulation.

For a first intuition, consider a feature as a vector in space (Fig. 1). The direction of the vector corresponds to its syntactic orientation (its basis), while its extent along each axis corresponds to its interface-relevant content (its components). [iF]s are vectors in the main feature space V, pointing in directions that are legible at CI. [uF]s, by contrast, inhabit the dual space V*; they are oriented “across” the main space and, as such, cannot contribute themselves contribute to the CI-interpretation – but can probe [iF]s.

Formally, a feature can be represented as a vector in the k-dimensional space V:

$$\mathbf{d}^{j} = \sum_{i=1}^{k} d^{ij}\mathbf{e}_{i} \tag{1}$$

where $\{\mathbf{e}_i\}$ are the basis vectors determining the syntactic dimensions of the space, and d^{ij} the components indicating the contribution of each dimension to the feature. A vector is thus characterised by its dimensionality k, its basis, and its components. I propose that these map onto syntactic and interface properties as follows: the choice of basis is syntactically relevant, the component values are interface-relevant, and dimensionality links the two.

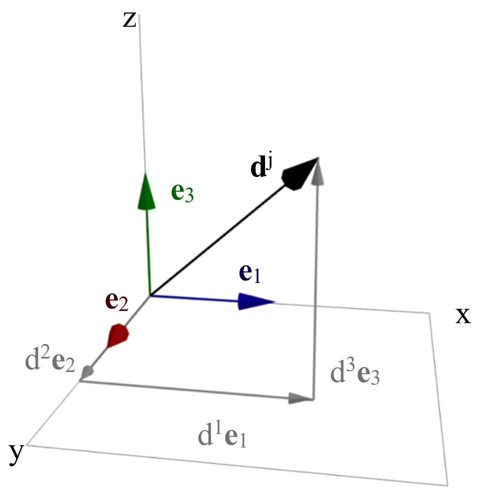

Fig. 1 The decomposition of a vector $\mathbf{d}^{j}$ in terms of its standard basis $\{\mathbf{e}_k\}$ and its components d^{ij}.

Within this framework, the distinction between [iF]s and [uF]s reduces to a distinction in orientation. [iF]s are vectors in the main space V, defined on a standard basis $\{\mathbf{e}_i\}$. [uF]s are covectors in the dual space V*, defined on the dual basis $\{\mathbf{e}^i\}$, which is orthogonal to V. Intuitively, [iF]s are *columns* that carry semantic content to CI, whilst [uF]s are *rows* that probe and act on [iF]s without interpretability at CI. Formally:

$$\begin{aligned} [\text{iF: Val}] &= \mathbf{d}^{j} = d^{ij}\mathbf{e}_i \\ [\text{uF: Val}] &= \mathbf{q}_j = q_{ij}\mathbf{e}^i \end{aligned} \tag{2}$$

An unspecified [uX] corresponds to a covector with no specific length along any axis; it consists solely of the dual basis, with no component values specified:

$$[\text{uX}] = \mathbf{q} = \sum_{i=1}^{k} \mathbf{e}^i \tag{3}$$

Intuitively, [uX] is a maximally underspecified probe, sensitive only to dimensional structure.

Checking consist in letting such a probe locate a suitable goal and mapping the result of that interaction into the main space. Crucially, the original convector does not itself become interpretable. Rather, it facilitates the creation of a new vector in V that reflects the outcome of the interaction. Checking is therefore modelled as the mapping from the dual space to the main space , $f: V^* \rightarrow V$, such that probing covector **q** gives rise to a new vector $\mathbf{p}^j = f(\mathbf{q})$ (Fig. 2). In these terms, AGREE corresponds to a change in basis induced by the probe-goal interaction.[2]

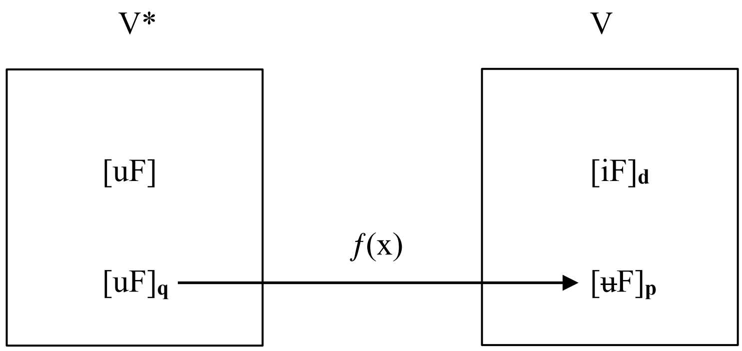

Fig. 2 Inter-dimensional mapping between two vector spaces: the standard vector space V and the dual vector space V*. All [uF]s belong to V*, whereas both [iF]s and checked [~~u~~F]s belong V. Mapping is implemented via a change in basis in the feature-vector representation.

Two immediate consequences follow. First, AGREE is inter-dimensional: covectors probe vectors, but vectors cannot probe covectors, reflecting the inherent directionality of the operation.[3] Second, AGREE is dimension-reducing: mapping from V* to V collapses uninterpretable structure into the space of interpretable features, triggering Transfer once all covectors have been so mapped.

The mapping *f* involves two components. The first is an interaction term, measuring how well a probing aligns with its potential goal. This is formally captured by the scalar product of a covector and vector:

$$\langle \mathbf{q}, \mathbf{d}^j \rangle = \mathbf{q}(\mathbf{d}^j) = d^j \tag{4}$$

This interaction measures dimensional alignment between probe and goal, reproducing the effect of the traditional Copy operation without literal copying: structural information is inherited insofar as the probe and goal share dimensional structure

The second component is a metric-like tensor g^{ij}, which implements the actual syntactic mapping from the dual space into the main space. This mapping is gated by an activation function σ, which ensures that transformation only proceeds when the interaction is satisfactory (in the sense of Deal's 2015 satisfaction condition). The interaction term is thus a necessary condition on mapping, while the metric supplies the structural means for mapping; only their conjunction, mediated by the gate, constitutes a sufficient condition for *f*.

Putting these elements together, the mapping can be schematically represented as:

$$[\cancel{u}F\text{: Val}] = \mathbf{p}^{j} = f(\mathbf{q}) = \sum_{i=1}^{k} \mathbf{q}\left(\mathbf{d}^{j}\right)^{i} \cdot \sigma\left(\mathbf{q}\left(\mathbf{d}^{j}\right)\right)^{i} \cdot g^{ij}\mathbf{e}_{i} \qquad (5)$$

Nothing in what follows hinges on the precise algebraic form of this expression, except for the fact that successful interaction introduces a non-linear mapping into V, a property that will be shown to be crucial for the invisibility of checked features at CI. Importantly, Valuation emerges as a by-product of successful interaction, rather than as an independently targeted operation. This account is compatible with the approaches developed in Bobaljik (2008), Kučerová (2018) and Epstein et al. (2022), which position Valuation post-syntactically.

The AGREE-procedure can now be restated in vectorial terms:

(7) AGREE for [uX] ∈ V*:	Probe	Goal
(i) Search (Probe): Search V (by MS).	[uX:__]	
(ii) Match: Locate a compatible [iF] and establish a coordinate map f: V* → V.	[uX:__]	[iF:Val]
(iii) Copy: Compute dimensionality alignment via the interaction term.	[uF:__]	[iF:Val]
(iv) Check: If interaction is satisfactory, apply the mapping f.	[~~u~~F:__]	[iF:Val]
(v) (Valuation): emerges from successful interaction.	[~~u~~F:Val]	[iF:Val]

In sum, the featural system can be understood geometrically: (un)interpretability is a property of representational orientation, and AGREE is an inter-dimensional search-and-mapping procedure. From here, the CI-filter follows directly from the internal architecture of feature representations, rather than being imposed as an independent interface condition.

3. CI-Filter: Null Space Hypothesis

A central property of the CI-interface is recoverability: under the Copy theory of Movement, Narrow Syntax builds syntactic objects via MERGE and transfers them to the interfaces. At the phase level, Form Copy (FC; cf. Chomsky et al. 2023) assigns the same interpretation to structurally identical copies. Given the duality of semantics (Chomsky, 2019; Chomsky et al., 2023), this entails that copies, and their semantic roles, must be relationally and invertibly recoverable: each interpreted object must be traceable to its original theta-position. This does not imply that MERGE itself is a linear operation, nor that CI admits only linear computations. It does imply, however, that CI-accessible representations must be recoverable in principle. Non-invertible distortions that destroy recoverability are therefore illicit at CI. SM, by contrast, imposes no such requirement; it merely requires that one copy be assigned an exponent, typically the highest one (though see Nunes 2004). As a result, non-recoverable or non-invertible distortions are admissible at SM, but excluded at CI.

This asymmetry naturally extends to the featural level. Consider the interaction term $\mathbf{q}(\mathbf{d}^j)$ introduced by AGREE. As long as this interaction remains linearly recoverable, CI can in principle interpret it. Once the non-linear transformation $\sigma(\mathbf{q}(\mathbf{d}^j))$ is applied, however, distinctions collapse: the mapping becomes many-to-one, and the original input cannot be uniquely reconstructed. Moreover, non-linearity affects not only the components values, but also the effective dimensional structure of the space by scaling the metric tensor. If, following Churchland (1993), dimensions themselves are semantically relevant, such deformation is catastrophic for CI-interpretation and must thus be filtered out.

I formalise this restriction via a Null Space CI-Filter, inspired by the Output-Potent/Output-Null distinction in factor-dynamics framework of neural computation (Churchland et al., 2012; Churchland & Shenoy, 2024; Kaufman et al., 2014; Perich et al., 2018). In this framework, activity in the Potent Space contributes directly to downstream computation, while activity int the Null Space does not. Translating this architecture to the linguistic domain, projection to the Potent Space corresponds to interpretability at CI. [iF]s project to the Potent Space; [~~u~~F] projects to the Null Space (Fig. 3).

Importantly, the Null Space is not computationally inert. In neural systems, Null Space dynamics redistribute and untangle the internal structure, increasing robustness and facilitating downstream readout (cf. Russo et al. 2018). I propose that the linguistic analogue is structural expansion: MERGE increases representational dimensionality, thereby untangling dependencies and enhancing linear recoverability. For example, when an argument bearing multiple features undergoes Internal MERGE, the resulting copies increase dimensionality while rendering the dependencies more transparent. FC can then linearly read out structurally linked copies (Chomsky et al., 2023), parallelling the well-known result that high-dimensional representations support linear readout (Fusi et al., 2016; Rigotti et al., 2010). This pattern is consistent with recent work arguing that linguistic representation are inherently high-dimensional (Chomsky, 2019b, 2019a, 2020, 2021; Pan & Du, 2024), and with approaches that reduce AGREE to FC (Roberts, 2024).[5] In this sense, FC crucially requires invertible recoverability and therefore corresponds to the projection of MERGEd structure onto the CI-Potent Space. Additionally, this proposal aligns with Chomsky's (2019b) reorientation

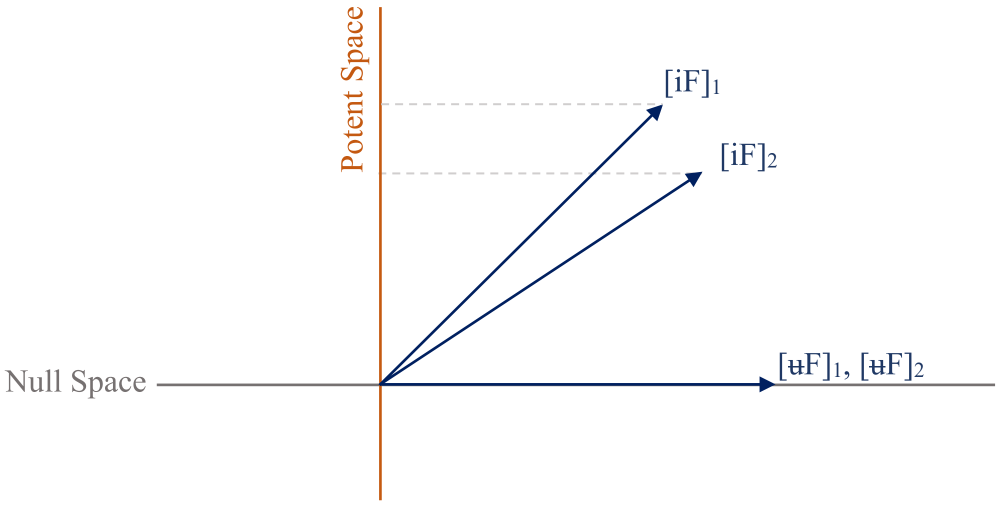

Fig. 3 Simplified[4] illustration of the Null Space CI-Filter Hypothesis. The CI-Filter comprises a Potent Space, which captures the contribution of the features to CI, and a Null Space, which captures features yielding zero CI-output. Checked [~~u~~F]s project onto null vectors, whilst [iF]s project onto the CI-Potent Space.

from External MERGE to Internal MERGE, insofar as it reconstrues the latter as a disentanglement operation rather than as an imperfection of the language system.

At the featural level, by contrast, AGREE introduces a non-invertible non-linearity. Since CI admits only invertibly recoverable structure, only linearly recoverable features project to the Potent Space. [iF]s survive; [~~u~~F], distorted by the non-linear mapping, collapse to a null vector and are thus *deleted*. At SM, where recoverability is irrelevant, both [iF]s and [~~u~~F]s are eligible for exponency.

This framework unifies CI-filtering across the phrasal and featural domains. CI enforces recoverability by projecting non-recoverable material into the Null Space. MERGE expands structure to enable linear readout under FC, while AGREE introduces non-linearities that force [~~u~~F]s into the Null Space. On this view, both FC and Deletion follow from a single CI-Filter, with interface conditions derived from the internal architectural constraints rather than imposed by stipulation.

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5. Footnotes

I want to thank Kim Groothuis for her support and supervision over the years, and Daniel Seely for his comments on earlier versions of this paper.

1. No claim is made that syntactic computation is literally vectorial; the vector formalism is adopted as an abstract representational metalanguage, intended to make explicit the architectural constraints that follow from recoverability and interface legibility. The substantive claim is that once these constraints are stated in vectorial terms, the CI-Filter follows as a consequence of internal architecture rather than stipulation.

2. It should be noted that Labelling and AGREE have been (partly) reduced to MS (Ke, 2019, 2022). Interestingly, the difference between the two operations within the current framework amounts to the space it is operative in; whilst Labelling

looks for features within the same vector space V, AGREE is operative inter-dimensionally from V* to V.

3. The flip mapping from V* to V under the hood is a mapping from V* to the double dual V**, where V** is isomorph to V. Despite this isomorphism, this identification is non-canonical and certainly ruled out under economy considerations.

4. For a more complex version of the Null Space Hypothesis, see, for instance, Churchland and Shenoy (2024).

5. This framework is also compatible with approaches that analyse Copying as a form of feature scattering rather than feature inheritance (cf. Erlewine, 2018; Gallego, 2014, 2017; Martinović, 2023). On such views, Copy need not involve literal transmission of feature values, but can be understood as the redistribution of featural information driven by checking requirements. Nothing in the present proposal hinges on this reinterpretation; the point is merely that modelling Copy via interaction and projection naturally aligns with a close architectural relation between AGREE and MERGE.